

JEE MAIN REVISION MASTER: ELECTROSTATICS

2026 5 QUESTIONS 8.33% WEIGHTAGE ↓28.62%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 53 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Electrostatics - Core Application	53 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Electrostatics - Core Application (Very High Yield — 53 PYQs)

Core Principle: Key problem pattern in Kcet Electrostatics.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A 200 J of work is done in moving a charge 5 C from a point A where the potential is -20 V to another point B where potential is V vol..." [Ans: N/A]
- [2026 Shift] Q#1 (MCQ): "An electron falls through a distance 1.5 cm in 2.0×10^4 N/C from rest. The time taken to cover this distance in second is ____ ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "An electric dipole of dipole moment $\vec{p}$ is placed in the uniform electric field $\vec{E}$. Then which of the following statements are..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A 200 J of work is done in moving a charge 5 C from a point A where the potential is -20 V to another point B where potential is V volt. The value of V at B is

- (A) 10 V
- (B) 20 V
- (C) 40 V
- (D) 60 V

Correct Answer: None

Solution Analysis: Work done in moving a charge from one point to another is related to potential difference by $W = q(V_B - V_A)$ Here, $W = 200\text{ J}$ $q = 5\text{ C}$ $V_A = -20\text{ V}$ So, $200 = 5[V_B - (-20)]$ $200 = 5(V_B + 20)$ Divide both sides by 5 : $40 = V_B + 20$ $V_B = 20\text{ V}$ So, the value of potential at point B is 20V Correct option: Option B.